

1045 Customers Who Bought All Products

Input

- Customer table — customer_id, product_key (may contain duplicate rows)
- Product table — product_key (primary key, list of all available products)

Output

- customer_ids who bought **every single product** listed in the Product table

Key Insights

- This is a **"bought everything" / set-coverage** problem
- Compare **count of distinct products per customer** to **total product count**
- Must use COUNT(DISTINCT ...) since Customer table may have duplicate rows
- Never hardcode the total product count — always use a subquery for it (dynamic & robust)

Observations

- Duplicate purchases of the same product shouldn't be counted twice → DISTINCT is essential
- A customer "qualifies" only if their distinct product count **equals** total products (not just ≥, since they can't buy more products than exist)
- This is conceptually similar to 570/1050 (GROUP BY + HAVING), but compares against a **dynamic subquery value** instead of a fixed number

Approach

1. GROUP BY customer_id → group all purchases per customer
2. COUNT(DISTINCT product_key) → count unique products each customer bought
3. HAVING compare this count to (SELECT COUNT(*) FROM Product) → total products available

Algorithm

- Step 1: GROUP BY customer_id
→ groups all purchases per customer
- Step 2: COUNT(DISTINCT product_key) per group
→ counts unique products bought
- Step 3: HAVING count = (SELECT COUNT(*) FROM Product)
→ keep only customers who bought ALL products

Time & Space Complexity

	Complexity
TC	O(N + M) — N = Customer rows, M = Product rows
SC	O(K) — K = number of distinct customers

Database

DescriptionEditorialSolutionsSubmissions

1045. Customers Who Bought All Products

Solved

MediumTopicsCompanies

SQL SchemaPandas Schema

Table: Customer

Column Name	Type
customer_id	int
product_key	int

This table may contain duplicates rows. customer_id is not NULL. product_key is a foreign key (reference column) to Product table.

Table: Product

Column Name	Type
product_key	int

product_key is the primary key (column with unique values) for this table.

Write a solution to report the customer ids from the Customer table that bought all the products in the Product table.

Return the result table in any order.

The result format is in the following example.

Example 1:

1.2K11526 Online

Accepted

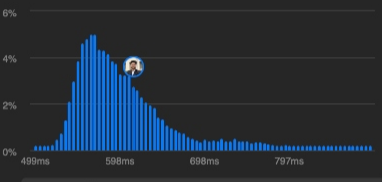
All Submissions

Accepted9 / 9 testcases passed

Sahil Khan submitted at Jun 29, 2026 13:03

Runtime

614 ms | Beats 37.35%



Code | MySQL

```
1 # Write your MySQL query statement below
2 SELECT customer_id
3 FROM Customer
4 GROUP BY customer_id
5 HAVING COUNT(DISTINCT product_key) = (SELECT COUNT(*) FROM product)
```

Write your notes here

Select related tags0/5

Code

MySQLAuto

```
1 # Write your MySQL query statement below
2 SELECT customer_id
3 FROM Customer
4 GROUP BY customer_id
5 HAVING COUNT(DISTINCT product_key) = (SELECT COUNT(*) FROM product)
```

SavedLn 1, Col 1

TestcaseTest Result

You must run your code first